Chapter 14, Problem 6RQ

Problem

In a series *RLC* circuit, which of these quality factors has the steepest magnitude response curve near resonance?

(a) $Q = 20$

(b) $Q = 12$

(c) $Q = 8$

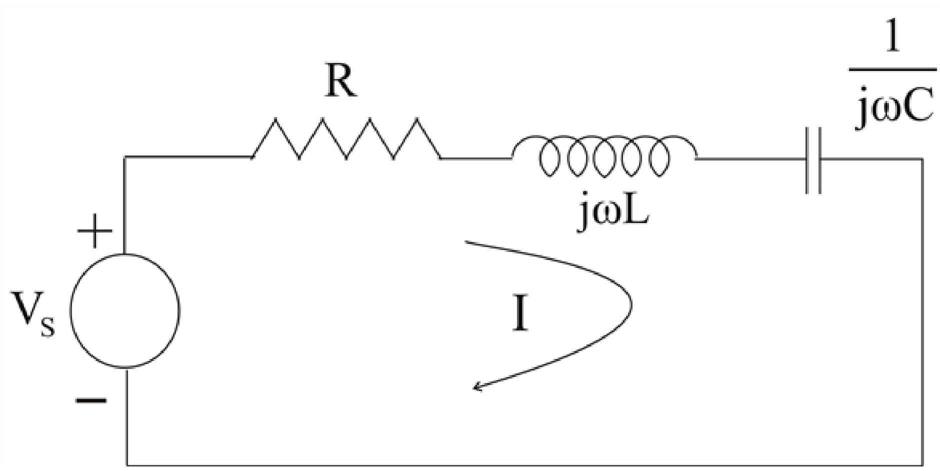
(d) $Q = 4$

Step-by-step solution

Step 1 of 4

The series resonant circuit is,

Step 2 of 4



Step 3 of 4

The input impedance is

$$Z = H(\omega) = \frac{V_S}{I} = R + j\omega L + \frac{1}{j\omega L} \quad -(1)$$

$$Or \quad Z = R + j\left(\omega L - \frac{1}{\omega E}\right) \quad -(2)$$

At resonance, ω=ω_o

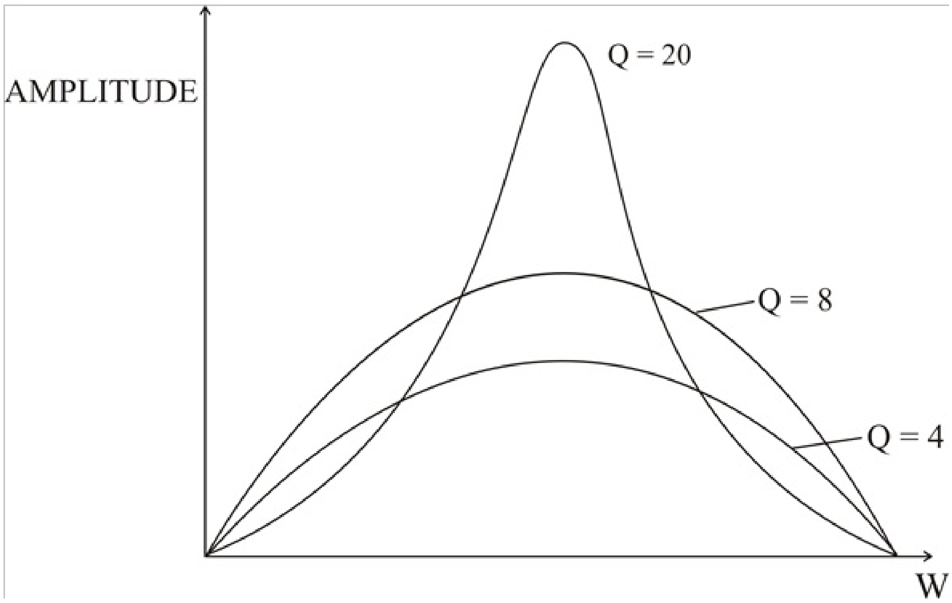
$$I_m = (Z) = 0.$$

$$\omega = \omega_o, \qquad \omega \Lambda - \frac{1}{\omega C} = 0$$

$$\Rightarrow \omega_o = \frac{1}{\sqrt{LC}} \quad -(3)$$

A Resonant circuit is said to be high quality if quality factor W equal to or greater than 10, $(\theta \geq 10)$.

There fore θ =20 has the steepest curve at resonance


AMPLITUDE
Q = 20
Q = 8
Q = 4
W